Radio Standards

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1. Introduction

Organization

Lecture
Takes place on Mondays, 03:00 pm - 04:30 pm, in OEC3008 (except winter break)

Tutorial

- Takes place on Tuesdays, 11:00 am - 12:30 pm, in OEC3008 (except winter break)
- No tutorial in the classical sense, instead three group presentations about selected standards
- Learn to understand standards and improve presentation skills
- Topic and group selection via moodle (deadline next lecture)
- DVB, 0G-5G, WiFi, and Bluetooth not selectable, will be covered in this lecture
- Feel free to suggest interesting standards by yourself

Prerequisites

- Signal and system theory
- Basic engineering math

Lecture Material

- Please enroll in the Moodle course
- Lecture script will grow throughout the semester

Exam Regulations

Exam

- **30 min** oral exam
- Standards presented by your groups will be part of the exam

Bonus Points

Presentations give bonus points for the exam

Maximum: One-third grade step

e.g. $2.7 \rightarrow 2.3$, $1.3 \rightarrow 1.0$

Overall Goal

How to work with standards

Understand why standards exist	How to read them	How to understand them (on technical level)	How to make use of them

Electronic Measurements and Signal Processing Group EMS

We investigate the behavior of radio propagation channels and develop methods to make them usable.





- Development of measurement equipment for characterizing radio channels from 0 Hz to 300 GHz
- Measurement, simulation, and modeling of radio channels
- Analysis of radio channel characteristics using signal processing - high-resolution estimation methods and AI

- Exploitation of radio channel characteristics for environmental sensing - radar technology and Integrated Communications and Sensing (ICAS)
- Emulation of electromagnetic wave propagation for testing mobile and satellite communications
- Standardization and development of solutions relevant to mobile communications standards

Radio Standards several decades ago...

... by Dr.-Ing. Alexander Ihlow

Topics to be Covered I

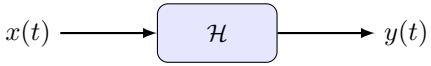
1. Technical Preliminaries
   - LTI Channel
   - Fourier Transformation
   - Spatial Processing
2. Propagation in a Nutshell (by Alexander Ihlow)
   - How do cable, satellite, and air interfaces differ from each other?
3. Standardization 101
   - How is the electromagnetic spectrum occupied?
   - What is a standard?
   - What is a standardization body?
   - What is a patent?
4. Evolution of Cellular Standards
   - 0G-2G
   - 3G
   - Deep dive OFDM
   - 4G
   - 5G

Topics to be Covered II

5. WiFi vs. Bluetooth vs. Cellular
   - Why do we have different standards for different applications and use cases?
6. Hands-on DVB (by Alexander Ihlow)
   - How to read a standard?
7. Business Models
   - How to commercialize a standard?
8. The Future of Mobile Communications
   - 6G and beyond
   - Trends and goals
   - Recent research at EMS

LTI System Model

Channel as Block Model



- System operator $\mathcal{H}$
- Transforms input signal $x(t)$ to output signal $y(t)$

System Operator under LTI Properties

- **L**inear
  - $\mathcal{H}\{\alpha x_1(t) + \beta x_2(t)\} = \alpha\mathcal{H}\{x_1(t)\} + \beta\mathcal{H}\{x_2(t)\}$
  - Superposition principle
- **T**ime **I**nvariant
  - $\mathcal{H}\{x(t-\tau)\} = y(t-\tau)$
  - Time delay τ of input equates to time delay τ of output

Remark: Real channels are inherently dynamic—covered in lecture 2 (coherence time)

Channel Impulse Response (CIR)

Response to an ideal Dirac pulse $\delta(t)$

$$h(t) = \mathcal{H}\{\delta(t)\}$$

LTI Input-Output Relation

Convolution of input $x(t)$ with CIR $h(t)$

$$y(t) = x(t) * h(t)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,\mathrm{d}\tau$$

Fourier Transformation I

From Impulse Response to Transfer Function

- Time-domain CIR $h(t)$ describes channel delay behavior
- Fourier Transform (FT) maps CIR to frequency domain Channel Transfer Function (CTF)
- $h(t) \circ\!\!-\!\!\bullet\, H(f)$

Basis Functions & Eigenfunction Property

- Complex exponentials $\phi_f(t) = \mathrm{e}^{\mathrm{j}2\pi ft}$ serve as orthogonal basis (kernel)
- Exponentials pass through unchanged except for complex scaling
- $\mathcal{H}\{\mathrm{e}^{\mathrm{j}2\pi ft}\} = H(f) \cdot \mathrm{e}^{\mathrm{j}2\pi ft}$
- Complex scalar $H(f)$ gives channel attenuation and phase rotation at frequency f

FT (Analysis)

Decomposition into spectrum

$$H(f) = \int_{-\infty}^{\infty} h(t)\, \mathrm{e}^{-\mathrm{j}2\pi ft}\, \mathrm{d}t$$

Inverse FT (IFT) (Synthesis)

Reconstruction of time signal

$$h(t) = \int_{-\infty}^{\infty} H(f)\, \mathrm{e}^{\mathrm{j}2\pi ft}\, \mathrm{d}f$$

Orthonormality of Complex Exponentials

$$\left\langle \mathrm{e}^{\mathrm{j}2\pi f_1 t}, \mathrm{e}^{\mathrm{j}2\pi f_2 t} \right\rangle = \delta(f_1 - f_2)$$

Fourier Transformation II

Time-Frequency Uncertainty

- Direct consequence of the FT scaling property
- $x(at) \circ\!\!-\!\!\bullet \frac{1}{|a|} X\left(\frac{f}{a}\right)$
- Narrow time duration $\implies$ wide spectral occupancy
- Time-limited signals can never be strictly band-limited

Impact on Radio Standards

- Sharp transitions (e.g., rect pulse) cause slow $1/f$ spectral decay
- High sidelobes leak energy thereby causing interference
- Regulatory spectral masks strictly limit out-of-band emissions

Remark: An optimal pulse must balance finite duration and low spectral sidelobes—covered in lecture 2 (root-raised-cosine)

FT Example

$$\mathrm{rect}\left(\tfrac{t}{T}\right) \circ\!\!-\!\!\bullet T\,\mathrm{si}(\pi f T)$$

$\mathrm{si}(x) = \frac{\sin(x)}{x}$, zeros at $f = \pm\frac{k}{T}$

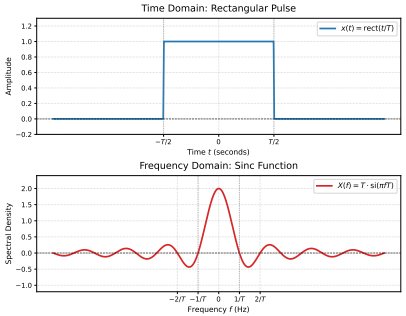


Abb.: Rectangular Pulse and Si Function

Fourier Transformation III

Live demo: Frequency spectrum from the computer microphone

- scripts/section-0/live-spectrogram.py

Fourier Transformation IV

Essential Fourier Properties

- Time Shifting (e.g. Propagation Delay)
  - $x(t-\tau_0) \;\circ\!\!-\!\!\bullet\; X(f)\,\mathrm{e}^{-\mathrm{j}2\pi f\tau_0}$
  - Delay τ_0 introduces a linear phase rotation across frequency
- Frequency Shifting (e.g. Doppler Shift)
  - $x(t)\,\mathrm{e}^{\mathrm{j}2\pi f_\mathrm{D} t} \;\circ\!\!-\!\!\bullet\; X(f-f_\mathrm{D})$
  - Doppler shift f_D introduces linear phase rotation across time
- Convolution Theorem
  - $x(t) * h(t) \;\circ\!\!-\!\!\bullet\; X(f)\cdot H(f)$
  - Simplifies channel equalization from deconvolution to division
- Sampling / Periodicity Duality
  - Periodicity t/f $\Longleftrightarrow$ Sampling f/t

$$\underbrace{x_\mathrm{s}(t) = x(t)\cdot\sum_{n=-\infty}^{\infty}\delta(t-nT_\mathrm{s})}_{\text{Sampling in Time}} \quad \circ\!\!-\!\!\bullet \quad \underbrace{X_\mathrm{s}(f) = \frac{1}{T_\mathrm{s}}\sum_{k=-\infty}^{\infty} X\left(f-\frac{k}{T_\mathrm{s}}\right)}_{\text{Periodification in Frequency}}$$

Complex Baseband I

Why Complex Signals for Real Waves?

- Antennas emit and capture real-valued signals $s_{\mathrm{BP}}(t) \in \mathbb{R}$
- Communications signals usually narrowband ($B \ll f_{\mathrm{c}}$):
  - Carrier f_{c} fixed in the GHz regime (e.g., $3.5\,\mathrm{GHz}$)
  - Information encoded in amplitude $A(t)$ and phase $\phi(t)$
- Simulating carrier oscillations at GHz rates is computationally expensive

Equivalent Complex Baseband Envelope

- Utilize complex numbers to combine amplitude and phase
- $s_{\mathrm{BB}}(t) = A(t)\,\mathrm{e}^{\mathrm{j}\phi(t)} \in \mathbb{C}$
- Describes signal independent of the carrier frequency

Passband Baseband Transformation

Passband as real projection of baseband signal

$$s_{\mathrm{BP}}(t) = \mathrm{Re}\Big\{ s_{\mathrm{BB}}(t)\,\mathrm{e}^{\mathrm{j}2\pi f_{\mathrm{c}} t} \Big\}$$

Advantages

- Spectrum centered around $0\,\mathrm{Hz}$
- Sample rate reduces from $2(f_{\mathrm{c}} + B/2)$ to B
- Clean mathematical framework for modulation and filtering

Complex Baseband II

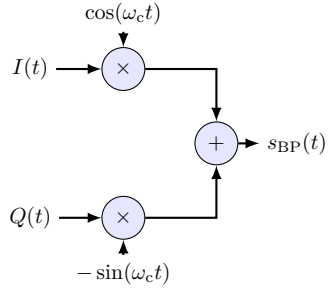
Cartesian Decomposition of Baseband

- Split complex envelope into real and imaginary parts:
- $s_{\mathrm{BB}}(t) = I(t) + \mathrm{j}Q(t)$
- $I(t) = \mathrm{Re}\{s_{\mathrm{BB}}(t)\} = A(t)\cos\phi(t)$ (**In-Phase**)
- $Q(t) = \mathrm{Im}\{s_{\mathrm{BB}}(t)\} = A(t)\sin\phi(t)$ (**Quadrature**)

Orthogonal Carrier Modulation

- Expanding $s_{BP}(t)$ with Euler's identity
- $\mathrm{e}^{\mathrm{j}2\pi f_{\mathrm{c}}t} = \cos\omega_{\mathrm{c}}t + \mathrm{j}\sin\omega_{\mathrm{c}}t$
- $s_{\mathrm{BP}}(t) = I(t)\cos(2\pi f_{\mathrm{c}}t) - Q(t)\sin(2\pi f_{\mathrm{c}}t)$
- Transmits two independent data streams over the same RF spectrum

I/Q Modulator Architecture



Why do we have Digital Communication Systems?

Disadvantages of Analog Systems

- Requires two physical signal traces for I and Q
- Analog hardware impairments

Advantages of Digital Systems

- Complex numbers handled in memory/software
- Flexibility of the hardware platforms (Software-defined radio)
- Conditional access and encryption
- Error correction and information compression
- Storage of digital data

Discrete Fourier Transformation I

Discrete Counterpart of the FT

- Discrete Fourier Transformation (DFT) maps N discrete time samples to N discrete frequency bins
- DFT kernel from FT kernel
- $\mathrm{e}^{-\mathrm{j}2\pi ft} \quad \longrightarrow \quad \mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}kn} = W_N^{kn}$
- Discrete complex exponentials $\phi_k[n] = \mathrm{e}^{\mathrm{j}\frac{2\pi kn}{N}}$ serve as orthogonal basis

Intrinsic Periodicity of the DFT

- Discrete harmonic basis is N-periodic
- $W_N^{(k+N)n} = W_N^{kn} \implies X[k+N] = X[k]$
- Time sequence implicitly treated as periodic
- $\tilde{x}[n] = x[n \bmod N]$

N-Point DFT (Analysis)

Time sequence to discrete frequency

$$X[k] = \sum_{n=0}^{N-1} x[n]\, \mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}kn}$$
$$k \in \{0, 1, \ldots, N-1\}$$

N-Point IDFT (Synthesis)

Reconstruction of time samples

$$x[n] = \tfrac{1}{N} \sum_{k=0}^{N-1} X[k]\, \mathrm{e}^{\mathrm{j}\frac{2\pi}{N}kn}$$
$$n \in \{0, 1, \ldots, N-1\}$$

Orthogonality of Discrete Complex Exponentials

$$\left\langle \mathrm{e}^{\mathrm{j}\frac{2\pi kn}{N}}, \mathrm{e}^{\mathrm{j}\frac{2\pi mn}{N}} \right\rangle = \begin{cases} N, & k = m \\ 0, & k \neq m \end{cases}$$

Discrete Fourier Transformation II

- Time Shifting
  - Compare to FT (Slide 1.12)
  - $x\left[(n - n_0) \bmod N\right] = X[k]\, e^{-j\frac{2\pi}{N}kn_0}$
- Frequency Shifting
  - Compare to FT (Slide 1.12)
  - $X\left[(k - k_0) \bmod N\right] = x[n]\, e^{j\frac{2\pi}{N}k_0 n}$
- Convolution Discrepancy Compared to FT
  - Discrete LTI channels perform linear convolution
  - $y[n] = x[n] * h[n] = \sum_m h[m]x[n-m]$
  - Multiplying two DFTs yields circular convolution ($\circledast_N$)
  - $x[n] \circledast_N h[n] \circ\!\!-\!\!\bullet\, X[k] \cdot H[k]$

Circular Convolution $\circledast_N$

$$x[n] \circledast_N h[n] = \sum_{m=0}^{N-1} h[m]\, x[(n-m) \bmod N]$$

Discrete Fourier Transformation III

DFT is a linear transformation

- N-point DFT acts as matrix-vector product:

$$\mathbf{X} = \mathbf{F}_N \mathbf{x}$$

- $\mathbf{x} = [x[0], \ldots, x[N-1]]^T \in \mathbb{C}^N$
- $\mathbf{X} = [X[0], \ldots, X[N-1]]^T \in \mathbb{C}^N$
- $\mathbf{F}_N$ is a symmetric matrix with entries $[\mathbf{F}_N]_{k,n} = W_N^{k \cdot n}$

Orthogonality & IDFT

- Columns of $\mathbf{F}_N$ form an orthogonal basis:

$$\mathbf{F}_N^H \mathbf{F}_N = N\mathbf{I}_N \implies \mathbf{F}_N^{-1} = \frac{1}{N}\mathbf{F}_N^H$$

- IDFT matrix is conjugate transpose $\mathbf{F}_N^H$ scaled by $1/N$

The DFT Matrix $\mathbf{F}_N$

$$\mathbf{F}_N = \begin{bmatrix} 1 & 1 & 1 & \ldots & 1 \\ 1 & W_N^1 & W_N^2 & \ldots & W_N^{N-1} \\ 1 & W_N^2 & W_N^4 & \ldots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & \ldots & W_N^{(N-1)^2} \end{bmatrix}$$

- With $W_N = \mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}}$

Spatial Processing I

- Plane wave = oscillation across distance and time
- Wave number k_0 = spatial phase velocity per meter $(\mathrm{rad/m})$
- Wave vector k = directional vector pointing along direction of propagation θ
- Spatial sampling = antenna elements sample positions as ADCs sample time
- Time–frequency duality maps to space–angle

Wave Basics

- Wave propagation along time and frequency
- $s(\mathbf{r}, t) = \mathrm{e}^{\mathrm{j}(\omega t - \mathbf{k}\cdot\mathbf{r})}$
- Spatial frequency vector
- $k_0 = \frac{2\pi}{\lambda}$

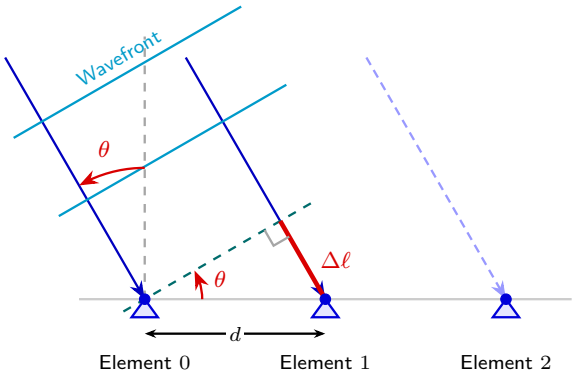
Time–Space Duality

Time Domain	**Spatial Domain**
Time t	Position x
Period T_s	Element Spacing d
Freq. ω	Wavenumber k_0
$f_\mathrm{s} \geq 2B$	$d \leq \lambda/2$

Spatial Processing II

Uniform Linear Array (ULA)

- Plane wave arriving from angle θ
- Wave travels extra Δl per element
- Elements experience angle-dependent phase shift ψ compared to reference element 0
- Array steering vector $\mathbf{a}(\theta)$
- $\mathbf{a}(\theta) = \begin{bmatrix} 1 & \mathrm{e}^{-\mathrm{j}\psi} & \dots & \mathrm{e}^{-\mathrm{j}(M-1)\psi} \end{bmatrix}^T$
- Spatial sampling without ambiguities if $d \leq \frac{\lambda}{2} \iff |\psi| \leq \pi$



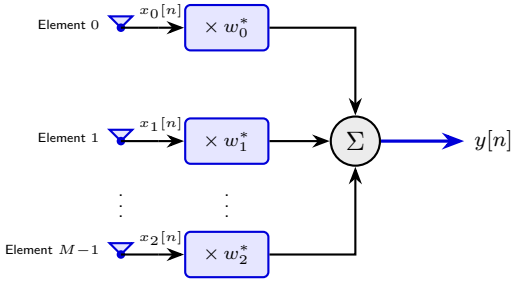
Path difference: $\Delta\ell = d \sin\theta$

Phase shift: $\psi = \frac{2\pi}{\lambda}\Delta\ell = \frac{2\pi d}{\lambda}\sin\theta$

Spatial Processing III

Bartlett Beamformer

- Weight vector $\mathbf{w}(\theta)$ "steers" antenna array
- Maximizes output power for direction θ by selecting weights as
- $\mathbf{w}_{\text{Bartlett}}(\theta) = \frac{1}{M}\mathbf{a}(\theta)$
- Undo the phase shifts introduced by the channel
- Scan output power of the beamformer across θ
- $P_{\text{BF}}(\theta) = \mathbf{w}^H(\theta)\,\mathbf{R}_{xx}\,\mathbf{w}(\theta)$



Output Signal Model

- Snapshot vector: $\mathbf{x}[n] = \mathbf{a}(\theta_0)s[n] + \mathbf{n}[n]$
- Beamformer output: $y[n] = \mathbf{w}^H(\theta)\mathbf{x}[n]$
- Spatial covariance: $\mathbf{R}_{xx} = \mathbb{E}\{\mathbf{x}[n]\mathbf{x}^H[n]\}$

Spatial Processing IV

Live demo: Bartlett beamformer visualization

- scripts/section-0/bartlett.py
- scripts/section-0/bartlett-multi-doa.py